

Sauhardha Kafle

+977 9847422110 [✉ Sauhardhakafle12@gmail.com](mailto:Sauhardhakafle12@gmail.com) [🌐 sauhardhakafle.com.np](http://sauhardhakafle.com.np) [🐙 github.com/sauhardh](https://github.com/sauhardh)

Skills

Languages: Rust, TypeScript, Python
Backend: Actix-web, NodeJS, ExpressJS, AIOHTTP, Flask, FastAPI
Frontend: NextJS, ReactJS, TailwindCSS,
Databases & Tools: Redis, MySQL, PostgreSQL, MongoDB, Appwrite
AI/ML: Deep Learning, Natural Language Processing, RAG
Infra: Docker, gRPC, IPC, webRTC, websocket

Experience

AI/LLM Fellowship | *KonnectCraft*

2026

Developed an ingestion pipeline for an graphical RAG application that extracts intelligence from the unstructured dataset. Also built an embeddable chatbot system enabling clients to integrate their custom RAG-powered assistants into their applications

Open Source Contributions

[Dora-rs](#) | *Rust*

- [PR #1341](#): Implemented global environment variables in dataflow YAML, allowing variables to be inherited by all nodes while preserving node-level overrides.
- [PR #1609](#): Fixed a pointer boundary check that was rejecting buffers starting at the beginning of a memory region.
- [Issue #56](#) – Identified and fixed an incorrect Conda environment name in the Dora documentation.

[St. Jude Rust Labs](#) | *Rust*

- [PR #9](#) – Replaced the deprecated `async-std` runtime with `smol`, updating integration, feature flags, and CI configuration.

Projects

[nutshell](#) | *TypeScript, Rust, Python, NextJS* | [Demo](#)

- A developer platform to build and deploy web application on cloud. A platform where we can host our website in few clicks similar to vercel.
- Uses **Cloudflare's R2** storage, **docker** to build the code, **redis** for information flow, **Rust** to handle the request, **NextJS** for frontend and **prisma postgres** for database. Completes overall hosting process within (~35 seconds.)

[CodeRoom](#) | *NodeJS, MySQL, Docker* | [Demo](#)

- A real-time collaborative coding platform with multi-user editing, chat, and live code compilation. A web-based **IDE** (Integrated Development Environment)
- Where I handled overall backend, **docker** to compile code which improved to (~0.2s) from regular API, **web socket** for real time messaging in **NodeJs**. Other feature includes **Authentication**, OTP code and uses **MySQL** for database.

[Torrex](#) | *Rust* | [Demo](#)

- Built a BitTorrent client from scratch in **Rust**, implementing Bencode-based .torrent parsing, magnet links via Extended Metadata Exchange, and HTTP tracker-based peer discovery.
- Implemented concurrent peer connections for piece downloading, **SHA-1** piece verification, and an **Actix Web** REST API with **WebSocket** streaming for real-time download progress.

[Aether](#) | *Python, Rust, WebRTC, WebSocket* | [Demo](#)

- A browser-based P2P remote desktop and compute-sharing platform enabling users to stream host screens and interact remotely in real time using **WebRTC** connections.
- Built a FastAPI signaling server using **Redis** Pub/Sub and **WebSockets**, with a **Next.js** WebRTC frontend, PostgreSQL, **NextAuth**, and Docker Compose.

Education

Kathmandu University

Dec 2022 – Feb 2027

Bachelor in Computer Engineering

Leadership & Activities

Technical Lead | *ITMeet*

2025

Led the technical team in developing AR-based marketing, an AR venue path finder, and the event website, while ensuring reliable internet infrastructure throughout the hackathon and venue.

Event Manager | *IdeaPitching*

2024

Technical Member | *ITMeet*

2024